

JAVIER REBULL, M.Sc.

Senior Full Stack Engineer | AI & ML Specialist | Software Architect

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Summary

Full stack engineer with 18 years delivering production systems in regulated financial services (credit cards, payments, banking), spanning end-to-end ownership from discovery and technical scoping through system design, build, and production rollout. Hands-on experience deploying machine learning models into production environments, building MLOps pipelines, and leading large-scale platform migrations without critical incidents. Known for aligning executive priorities, engineering teams, risk partners, and external banking institutions around measurable outcomes — and for keeping delivery moving in complex, high-stakes environments.

Work Experience

Sr. Associate Application Development

Mar 2024 – Present

Santander Bank US

Quincy, MA

- Led the Gravity Middleware Project end to end, migrating from IBM Mainframe/DB2 to private cloud (Microfocus/Oracle) without critical incidents, ensuring seamless operational continuity for production banking systems.
- Established end-to-end MLOps pipelines for automated model deployment and monitoring, driving operational efficiency in production banking environments.
- Directed full stack development for Certificate of Deposits at OpenBank, building user-friendly front-end interfaces and robust back-end services that improved customer engagement.
- Managed the transition of 500+ WAS/WebSphere servers to modern cloud infrastructure, architecting a scalable distributed system for performance and reliability.
- Conducted design and code reviews across engineering teams, codifying best practices in testability and code quality.

Associate Application Development

Sep 2019 – Feb 2024

Santander Bank US

Quincy, MA

- Led full stack development on high-pressure, deadline-driven initiatives including the SBNA Loans COVID-19 Relief Program and Regulatory FDIC 370, delivering on time and in compliance.
- Designed and developed UI/UX interfaces for internal banking applications, driving a 35% increase in user satisfaction and engagement.
- Diagnosed and resolved production issues by analyzing root causes across hardware, network, and service layers, improving uptime and performance.
- Maintained stable, incident-free production environments across distributed banking systems.

Computer Systems Analyst

Apr 2013 – Sep 2019

Vass IT US

Cambridge, MA

- Developed Computer Vision solutions for document processing and fraud detection, integrating machine learning models into client production systems.
- Managed full stack application development for loan management, leading IT project portfolios and PMO activities across cross-functional, client-facing teams to deliver on time and within budget.
- Oversaw software product development from requirements gathering through deployment and maintenance, ensuring alignment with client business objectives.

Spec Analytics Analyst 2

Jun 2010 – Apr 2013

Citibank Banamex

Mexico City, Mexico

- Designed and optimized secure distributed systems for global financial services, processing millions of daily transactions with high availability.
- Developed data-driven algorithms (SAS/Python) that increased marketing response rates by 20% and reduced delinquency by 10%.
- Built interactive dashboards and front-end interfaces for executive reporting and real-time data visualization.

Sr. Programmer

Aug 2007 – May 2010

Accenture

Mexico City, Mexico

- Developed and maintained large-scale core financial systems (COBOL, Java) for client banks BBVA Bancomer and Santander México, with zero critical incidents.
- Designed front-end user interfaces for mainframe applications, improving operational efficiency and developer experience.

Education

M.S. in Artificial Intelligence and Data Analytics

Universidad Autónoma de Ciudad Juárez (UACJ)

2025 – 2027 (Expected)

Ciudad Juárez, Mexico

Focus: Smart Optimization, Linear Optimization, Non-Linear Optimization, Data Engineering

M.Sc. in Applied Artificial Intelligence

Tecnológico de Monterrey (ITESM)

2024 – 2026 • Degree conferred

Mexico City, Mexico

GPA: 4.0/4.0 • Coursework: Data Science, Machine Learning, AI, Data Visualization, Fintech & IoT

B.S. in Computer Science & Electronics Engineering

The National Polytechnic Institute of Mexico (ESCOM IPN)

2002 – 2006

Mexico City, Mexico

GPA: 3.28/4.0 • Coursework: Programming, Software Engineering, AI, Computer Networks, Data Structures, Algorithms & Operating Systems

AI / ML Projects

- **EpiForecast-MX — Epidemiological Forecasting Platform** (2025–2026): M.Sc. capstone at Tecnológico de Monterrey, built with the Instituto Mexicano del Seguro Social (IMSS). Forecasts weekly notified cases across Mexico's 32 federal entities on a 52-week horizon, training Prophet, DeepAR, Ensemble and Stacking engines per series and selecting one through an auditable rule that is revised against each new national epidemiological bulletin. In production at epiforecast.mx.
- **Text Classification with RNNs — AG NEWS Dataset** (Mar 2025): Built an RNN in PyTorch to classify AG NEWS articles, covering data preprocessing, model architecture design, training pipelines, and evaluation.
- **IoT for Sustainable Urban Development** (Feb 2025): Investigated IoT solutions for traffic congestion in Boston and public lighting efficiency in Mexico City using distributed sensor networks.
- **Multilayer Fully Connected Network in NumPy** (Jan 2025): Implemented a neural network from scratch for ASL letter recognition, including forward/backward propagation and performance benchmarking.
- **Stock Price Prediction for Apple** (Nov 2024): Developed and compared LSTM, ARIMA, Prophet, and Naïve predictive models on historical market data using advanced statistical methods.

Conferences & Publications

- **ePoster — 26th WPA World Congress of Psychiatry**, Stockholm, Sweden (Sep 23–26, 2026), Digital Psychiatry track: “*Population-level prediction of depression, Alzheimer's disease, and Parkinson's disease incidence using artificial intelligence: implications for health service planning.*” With J. C. Pérez-Nava, L. G. Sánchez-Salazar, G. Barceló-Alonso, L. Díaz-Castro, S. M. Cuadra-Hernández, and R. M. Pérez-Hernández.
- **Oral presentation — CALASS 2026, Congrès de l'ALASS**, Université de Montréal, Canada (Aug 27–29, 2026): “*De la surveillance épidémiologique à l'intelligence prédictive : le rôle de l'intelligence artificielle dans l'organisation prospective des systèmes publics de santé.*” Presented EpiForecast-MX (Tecnológico de Monterrey & IMSS). With J. C. Pérez-Nava, L. G. Sánchez-Salazar, G. Barceló-Alonso, R. Pérez-Hernández, and S. M. Cuadra-Hernández.
- **Poster — 2nd Public Health Conference & 2nd HSG Europe Preconference**, Skopje, North Macedonia (May 21–22, 2026): “*AI-Based Forecasting to Shift Epidemiological Surveillance toward Predictive Approaches: Depression, Alzheimer's, and Parkinson's as Use Cases.*” Same co-authors as the WPA ePoster.
- **Forecasting Dengue in Mexico: An End-to-End Data Science Story** — Medium. End-to-end data science article covering the full workflow for forecasting dengue cases in Mexico.

Certificates: rebull.org/constancias

Training & Certifications

Python Essentials 1 & 2 — Cisco (Jul 2026) • Lifelong Learning — Certiprof (Jan 2026) • AWS Certified AI Practitioner (Dec 2025) • Scrum Master Professional — SMPCC® (Jun 2025) • Generative AI in Action — IBM SkillsBuild (Jun 2025) • AWS Certified Cloud Practitioner (Jan 2025)